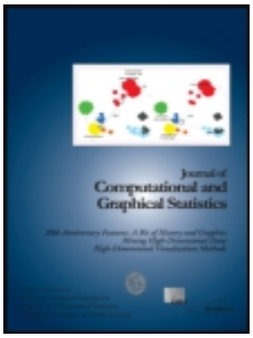

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Mixtures of Gamma Distributions With Applications

Michael WIPER, David RIOS INSUA, and Fabrizio RUGGERI

This article proposes a Bayesian density estimation method based upon mixtures of gamma distributions. It considers both the cases of known mixture size, using a Gibbs sampling scheme with a Metropolis step, and unknown mixture size, using a reversible jump technique that allows us to move from one mixture size to another. We illustrate our methods using a number of simulated datasets, generated from distributions covering a wide range of cases: single distributions, mixtures of distributions with equal means and different variances, mixtures of distributions with different means and small variances and, finally, a distribution contaminated by low-weighted distributions with different means and equal, small variances. An application to estimation of some quantities for a M/G/1 queue is given, using real E-mail data from CNR-IAMI.

Key Words: Bayesian inference; MCMC; M/G/1 queues; Reversible jump.

1. INTRODUCTION

There has been much recent work on the use of mixtures of normal distributions in Bayesian, nonparametric density estimation. See, for example, Diebolt and Robert (1994), Escobar and West (1995), Richardson and Green (1997) and Roeder and Wasserman (1997). However, there are situations in which it may be preferable to consider alternative families of mixture distributions; for example, when the intrinsic range of the variable is not the whole real line, or when the data being modeled exhibit skewness as well as multimodality. Here we consider inference for data defined over the positive reals $(0, \infty)$.

One possibility would be to consider a mixture of exponential distributions (or hyper-exponential distribution). This was considered by Gruet, Philippe, and Robert (1999) and, in a queueing context, by Rios Insua, Wiper, and Ruggeri (1998). However, this family is inappropriate for general density estimation, as it is easy to show that the coefficient of variation of the hyperexponential distribution is always greater than or equal to one, which does not of course hold in general.

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This article considers mixtures of gamma distributions of the form

$$f(x|k, \mathbf{w}, \boldsymbol{\mu}, \boldsymbol{\nu}) = \sum_{i=1}^{k} w_i \mathcal{G}(x|\nu_i, \nu_i/\mu_i),$$

where

$$\mathcal{G}(x|\nu_i, \nu_i/\mu_i) = \frac{\left(\nu_i/\mu_i\right)^{\nu_i}}{\Gamma(\nu_i)} x^{\nu_i - 1} \exp\left(-\frac{\nu_i}{\mu_i} x\right).$$

Mixtures of gamma distributions were also considered by Dey, Kuo, and Sahu (1995), though their parameterization is somewhat different than ours, and they consider only the case of a known mixture size. Note that our parameterization of the gamma densities is in terms of its mean μ_i and shape parameter ν_i.

By increasing the number of terms k in the mixture we can gain an arbitrarily close approximation to a general, absolutely continuous distribution on $(0, \infty)$. DeVore and Lorentz (1993, p. 14) showed for any continuous function $f(x)$ on $(0, \infty)$, with limit zero as $x \to \infty$, that $\lim_{u\to\infty} S_u(x) = f(x)$ uniformly for $0 < x < \infty$, where $S_u(x)$ is defined by

$$S_u(x) := e^{-ux} \sum_{k=0}^{\infty} \frac{(ux)^k}{k!} f\left(\frac{k}{u}\right), \quad u > 0.$$

Therefore, f can be approximated by a mixture of Gamma densities $\mathcal{G}(k+1, u)$, with weights $(1/u)f(k/u)$. In a Bayesian framework, Dalal and Hall (1983) showed how to approximate prior densities by mixtures of natural conjugate priors; in particular, they showed that any prior on the mean of a Poisson model can be approximated by an infinite mixture of Gamma densities $\mathcal{G}(k+1, K)$, with k and K both integers. Finally, Asmussen (1987, p. 76, theorem 6.2) showed that mixtures of Erlang distributions converge (weakly) to any probability measure on $(0, \infty)$.

Our general family of gamma mixtures is actually bigger than the one used in the previously mentioned results because the first parameter of the gamma distribution is not restricted to be an integer. As a consequence, we should typically obtain more parsimonious approximations than those provided above.

Section 2 considers Bayesian inference schemes conditional on the mixture size k and Section 3 uses a reversible jump algorithm (Green 1995; Richardson and Green 1997) to extend the modeling to the situation when k is unknown. Section 4 illustrates the method with various simulated examples and Section 5 introduces the specific problem of inference for the M/G/1 queue. There we use the methods introduced in this article to undertake inference concerning the equilibrium properties of the queue. Section 6 gives conclusions and possible extensions.

2. BAYESIAN INFERENCE CONDITIONAL ON THE MIXTURE SIZE k

2.1 THE PRIOR MODEL

We now suppose that given a sample of data $x_1, \ldots, x_n$, we wish to undertake inference for the parameters of the mixture model. Suppose initially that the number of elements of

the mixture k is known. Then, following Diebolt and Robert (1994), we introduce indicator variables $Z_1, \ldots, Z_n$ such that

$$P(Z_i = z|k, \mathbf{w}) = w_z \quad \text{for} \quad z = 1, \ldots, k.$$

Then, conditional on these indicators, we have

$$X_i|Z_i = z \sim \mathcal{G}(\nu_z, \nu_z/\mu_z).$$

For the remaining model parameters, we assume proper but relatively diffuse distributions of the form

$$\begin{array}{rcll} \mathbf{w}|k & \sim & \mathcal{D}(\boldsymbol{\phi}) & \text{a Dirichlet distribution} \\ \nu_z|k & \sim & \mathcal{E}(\theta) & \text{an exponential distribution, for} \quad z = 1, \ldots, k \\ \mu_z|k & \sim & \mathcal{GI}(\alpha, \beta) & \text{an inverted gamma distribution, for} \quad z = 1, \ldots, k \end{array}$$

with the restriction $\mu_1 < \cdots < \mu_k$.

Typically, we might set $\boldsymbol{\phi} = \mathbf{1}$ to give a uniform prior over the weights, $\theta = .01$, $\alpha = 1$, and $\beta = 1$. Note that the usual improper reference priors are inappropriate here, leading to improper posteriors; see Roeder and Wasserman (1997). Note that given $\alpha \leq 1$, the prior moments of $\boldsymbol{\mu}$ do not exist. Thus, it is also easy to prove that, for $k > 1$, the posterior moments do not exist. If we are interested in estimating the mixture parameters rather than just modeling the distribution of the data, it may well be more appropriate to use a larger value of α. We do this in Section 5 when we consider inference in a queueing scenario.

The order restriction on the prior of $\boldsymbol{\mu}$ is used to ensure identifiability of the posterior distribution although it has the disadvantage of, theoretically, not allowing us to model mixtures of distributions with equal means. In practice, see Section 4, this has not proved to be a problem. We could use instead a lexicographic ordering procedure, supposing that $\mu_j \leq \mu_{j+1}$ and that if $\mu_j = \mu_{j+1}$, then $\nu_j < \nu_{j+1}$, but inference under this scheme would be somewhat harder to implement and, for practical purposes, would seem unlikely to improve the estimates very much. We note that the identifiability restriction is unnecessary if our interest is solely in modeling the distribution of the observable X.

2.2 POSTERIOR INFERENCE

The conditional posterior distributions of the relevant model parameters given k are easily shown to be as below.

$$P(Z_i = z|\mathbf{x}, k, \ldots) \propto w_z \frac{\left(\nu_z/\mu_z\right)^{\nu_z}}{\Gamma(\nu_z)} x_i^{\nu_z - 1} \exp\left(-\frac{\nu_z}{\mu_z} x_i\right),$$

$$\mathbf{w}|\mathbf{x}, k, \ldots \sim \mathcal{D}(\boldsymbol{\phi} + \mathbf{n}),$$

where $n_z = \#\{Z = z\}$; for $z = 1, \ldots, k$ and $\sum_{z=1}^{k} n_z = n$

$$\mu_z|\mathbf{x}, k, \ldots \sim \mathcal{GI}(\alpha + n_z\nu_z, \beta + S_z\nu_z),$$

where $S_z = \sum_{i:Z_i=z} x_i$

$$f(\nu_z|\mathbf{x}, k, \ldots) \propto \frac{\nu_z^{n_z \nu_z}}{\Gamma(\nu_z)^{n_z}} \exp\left(-\nu_z \left(\theta + \frac{S_z}{\mu_z} + n_z \log \mu_z - \log P_z\right)\right), \tag{2.1}$$

where $P_z = \prod_{i:Z_i=z} x_i$.

Thus, conditional on k, we can define a Gibbs sampling algorithm to sample the full posterior distribution as follows:

1. $t = 0$. Set initial values $\mathbf{w}^{(0)}, \boldsymbol{\mu}^{(0)}, \boldsymbol{\nu}^{(0)}$.
2. Generate $\mathbf{z}^{(t+1)} \sim \mathbf{z}|\mathbf{x}, k, \mathbf{w}^{(t)}, \boldsymbol{\mu}^{(t)}, \boldsymbol{\nu}^{(t)}$.
3. Generate $\mathbf{w}^{(t+1)} \sim \mathbf{w}|\mathbf{x}, k, \mathbf{z}^{(t+1)}, \boldsymbol{\mu}^{(t)}, \boldsymbol{\nu}^{(t)}$.
4. Generate $\mu_z^{(t+1)} \sim \mu_z|\mathbf{x}, k, \mathbf{z}^{(t+1)}, \mathbf{w}^{(t+1)}, \boldsymbol{\nu}^{(t)}$ for $z = 1, \ldots, k$.
5. Generate $\nu_z^{(t+1)} \sim \nu_z|\mathbf{x}, k, \mathbf{z}^{(t+1)}, \mathbf{w}^{(t+1)}, \boldsymbol{\mu}^{(t+1)}$ for $z = 1, \ldots, k$.
6. Order $\boldsymbol{\mu}^{(t+1)}$ and sort $\mathbf{w}^{(t+1)}$ and $\boldsymbol{\nu}^{(t+1)}$ accordingly.
7. $t = t + 1$. Go to 2.

The only complicated step is that of sampling the distribution of ν_z, for $z = 1, \ldots, k$. It is easy to show, however, that for large ν_z, the density given by Equation (2.1) is similar to a gamma density, and thus we use a Metropolis step with a gamma candidate distribution to generate candidate values.

For each $z = 1, \ldots, k$ we generate a candidate $\tilde{\nu}_z \sim \mathcal{G}(r, r/\nu_z^{(t)})$ and accept this with probability

$$\min\left\{1, \frac{f(\tilde{\nu}_z|\ldots)p\left(\tilde{\nu}_z, \nu_z^{(t)}\right)}{f(\nu_z^{(t)}|\ldots)p\left(\nu_z^{(t)}, \tilde{\nu}_z\right)}\right\},$$

where $p(\tilde{\nu}, \nu)$ is the gamma density used to generate $\tilde{\nu}$ and $f(\tilde{\nu}_z|\ldots)$ is as in Equation (2.1). If the candidate is rejected, we retain the current value. The parameter r of the gamma proposal distribution may be adjusted to give a satisfactory acceptance rate. In practice, although we make no claims of universal efficiency, we have found that $r = 2$ has worked well.

3. INFERENCE WHEN k IS UNKNOWN

Suppose now that k is unknown and we define a prior density $P(k)$ with support $1, 2, \ldots, k^{\max}$. Typically, we choose $k^{\max} < n$ although, in theory, the support for k could be infinite when we would have full semi-parametric modeling. Obviously, if given data, the posterior probability that k is equal to $k^{\max}$ is high, this would suggest that $k^{\max}$ should be increased.

In practice, we have considered discrete uniform, truncated geometric and truncated, shifted Poisson distributions. Then we can extend the Gibbs sampling algorithm given in Section 2 by introducing a reversible jump to allow us to move through the posterior distribution of k. This is basically a Metropolis step which allows for a change in the

dimension of the parameter space. See Green (1995) for a general description of reversible jump methods and Richardson and Green (1997) in the context of mixtures of normals.

We modify the Gibbs algorithm of Section 2 by replacing k with $k^{(t)}$ throughout and rewriting Step 7 as

7. Generate $k^{(t+1)}$ and modify the other model parameters appropriately.
8. $t = t + 1$. Go to 2.

The reversible jump algorithm proceeds as follows. First, given a current value of k we generate a candidate $\tilde{k}$:

$$\tilde{k} = \begin{cases} k+1 & \text{if} \quad k = 1 \\ k-1 & \text{if} \quad k = k^{\max} \\ k-1 \quad \text{with} \quad p = .5 \quad \text{or} \quad k+1 \quad \text{with} \quad p = .5 & \text{if} \quad 1 < k < k^{\max} \end{cases}.$$

Given $\tilde{k}$ we now modify the other model parameters appropriately.

If $\tilde{k} = k - 1$, we must reduce the mixture size and thus we elect two neighboring terms in the mixture ($j1, j2 = j1 + 1$) to combine, at random, with probability $p = 1/(k-1)$. Supposing that $\tilde{w}$, $\tilde{\mu}$, and $\tilde{\nu}$ are the parameters of the combined term, we define:

1. $\tilde{w} = w_{j1} + w_{j2}$,
2. $\tilde{w}\tilde{\mu} = w_{j1}\mu_{j1} + w_{j2}\mu_{j2}$, and
3. $\tilde{w}\tilde{\mu}^2\left(1 + \frac{1}{\tilde{\nu}}\right) = \sum_{i=1}^{2} w_{ji}\mu_{ji}^2\left(1 + \frac{1}{\nu_{ji}}\right)$.

Using these transformations, we preserve the restriction that the weights sum to 1 and the conditional moments

$$E[X^i|k, \ldots] = E[X^i|\tilde{k}, \ldots]$$

for $i = 1, 2$.

The algorithm for increasing k to $k + 1$ is partially determined by the previous step. We now elect an element j of the mixture to split into two, with probability $1/k$. We then split the parameters as follows.

1. Generate $u_1 \sim U(0, 1)$ and define

$$\tilde{w}_{j1} = u_1 w_j, \quad \tilde{w}_{j2} = (1 - u_1)w_j$$

2. Generate $u_2 \sim U(0, 1)$ and define, for $j \geq 2$,

$$\begin{aligned} \tilde{\mu}_{j1} &= \mu_{j-1} + u_2(\mu_j - \mu_{j-1}) \\ \tilde{\mu}_{j2} &= \frac{(1 - u_1u_2)\mu_j - u_1(1 - u_2)\mu_{j-1}}{1 - u_1}, \end{aligned}$$

   where the formula when $j = 1$ may be derived by setting $\mu_0 = 0$ in the above.
3. Generate $u_3 \sim \mathcal{G}(s, s)$ and $\tilde{\nu}_{j1} = u_3\nu_j$ and $\tilde{\nu}_{j2}$ in consequence to satisfy Step 3 of the combination algorithm above. The parameter s can be adjusted by trial and error in order to achieve a satisfactory acceptance rate. Typically we have used $s = 5$.

Given the generated parameter values, we now choose whether to accept or reject the move. We note that in Steps 2 and 3 of the splitting algorithm, we may produce impossible

values of $\tilde{\mu}_{j2}$ and $\tilde{\nu}_{j2}$ and if this should happen, we reject the move from k to $\tilde{k}$ immediately. Otherwise, we accept it with probability

$$\min\left\{1, \frac{f(\tilde{\boldsymbol{\theta}})p_{\bar{m}}(\tilde{\boldsymbol{\theta}}, \boldsymbol{\theta})}{f(\boldsymbol{\theta})p_m(\boldsymbol{\theta}, \tilde{\boldsymbol{\theta}})}\right\},$$

where $p_m(\boldsymbol{\theta}, \tilde{\boldsymbol{\theta}})$ is the density of the move from $\boldsymbol{\theta}$ to $\tilde{\boldsymbol{\theta}}$ and $f(\boldsymbol{\theta})$ is the distribution of $\boldsymbol{\theta}$ given by

$$f(\boldsymbol{\theta}) \propto k! \left[\prod_{i=1}^{n}\left(\sum_{j=1}^{k} w_j \mathcal{G}(x_i|\nu_j, \nu_j/\mu_j)\right)\right] f(k, \mathbf{w}, \boldsymbol{\mu}, \boldsymbol{\nu}). \tag{3.1}$$

The densities of the moves are

$$p_{\text{combine}} = \frac{1}{2}\frac{1}{k-1}$$

and

$$p_{\text{split}} = \frac{1}{2}\frac{1}{k} f(u_3) \times \text{inv},$$

where $f(u_3)$ is the gamma density used to generate u_3 and inv is the Jacobian of the inverse transformation.

This algorithm is similar to that given by Richardson and Green (1997) for normal mixtures. The main difference is that we do not redefine the indicator variables Z_i, within the reversible jump algorithm. Note that this is done at the very next step of the Gibbs sampling algorithm and therefore is not a problem. The cost of this is that the density given in Equation (3.1) is more complicated, including the full mixture likelihood. We have experimented with both approaches and in practice and results and computing times have been approximately equal.

Given the MCMC output, we can estimate (for example) the predictive density of the data using

$$f(x|\mathbf{x}) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{k^{(i)}} w_j^{(i)} \mathcal{G}\left(x|\nu_j^{(i)}, \nu_j^{(i)}/\mu_j^{(i)}\right),$$

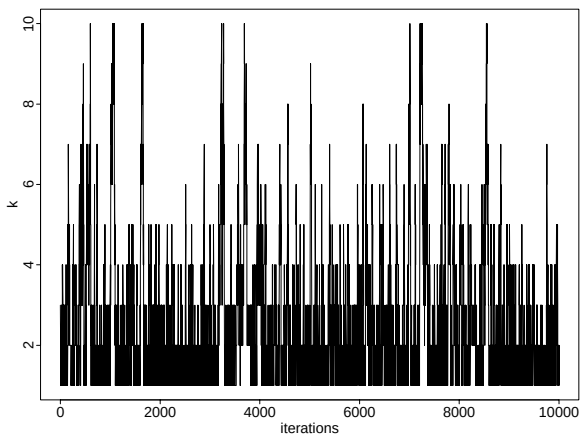
where $k^{(i)}, \ldots$ are the values generated assuming equilibrium. We can estimate densities of functions of X similarly. For example, if $Y = \log X$, then

$$f_Y(y|\mathbf{x}) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{k^{(i)}} w_j^{(i)} e^y \mathcal{G}\left(e^y|\nu_j^{(i)}, \nu_j^{(i)}/\mu_j^{(i)}\right).$$

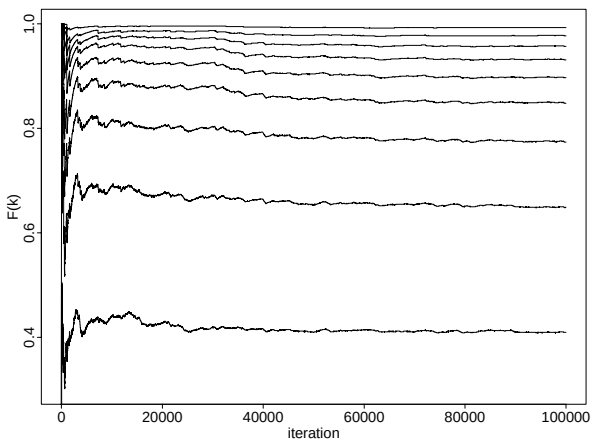
Sections 4 and 5 illustrate the performance of the algorithm with some simulated and real examples.

4. EXAMPLES

We consider simulated data from mixtures of gamma distributions covering a variety of cases in the spirit of Marron and Wand (1992), whose test functions were mixtures of normal distributions.



(a)



(b)

Figure 1. Data from $\mathcal{G}$(3,3). (a) Changes in k against iterations; (b) convergence of the estimated probabilities for $P(k \leq i)$ (i = 1, . . ., 9) against iterations.

4.1 A SINGLE DISTRIBUTION

Four hundred data were simulated from $\mathcal{G}(3, 3)$. We assumed a discrete uniform prior on $\{1, \ldots, 10\}$ for k and used the prior densities given in Section 2 for the other variables. A single Gibbs run of 100,000 iterations to burn-in and 100,000 iterations in equilibrium was used.

Figure 1(a) illustrates the changes in the value of k against iterations. The chain appears to be mixing quite well, visiting almost the whole of the support of the distribution for k. In fact, the overall proportion of moves in k accepted was around 10%.

Figure 1(b) illustrates the convergence of the estimated probabilities for $P(k \leq i)$ ($i = 1, \ldots, 9$) against iteration number. We can see that these estimated probabilities appear to have converged after about 50,000 iterations.

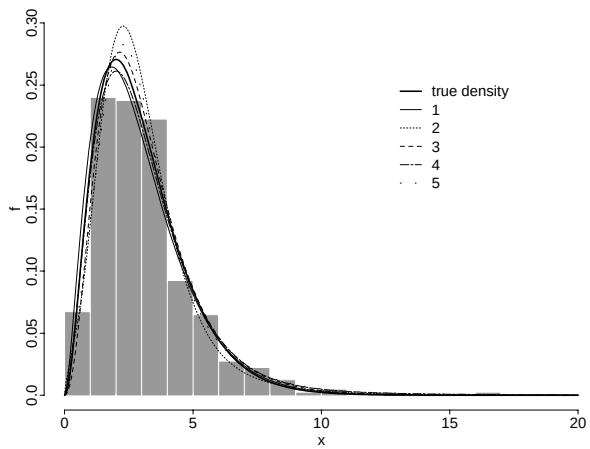


Figure 2. Densities sampled at various iterations.

We note that the estimated posterior probabilities for the different values of k were (approximately) $P(k = 1) = .41$, $P(k = 2) = .24$, $P(k = 3) = .12$, $P(k = 4) = .08$, $P(k = 5) = .05$ with other values of k having posterior probabilities less than .03. Thus, although there is some uncertainty here, the single gamma distribution appears to have been spotted by the method.

Figure 2 shows a histogram of the data, with the true generating density (bold solid line) and a single density selected at random from the MCMC sample for each of $k = 1, \ldots, 5$. We can see that the density estimates are all very similar in this case.

Finally, note that conditional on the mixture size k, we can also estimate the remaining model parameters. In this example, we find that given $k = 1$, the posterior mean of μ (which does exist in this case) is approximately 3.1 and the posterior mean for ν is 2.87. Thus, we observe that given the correct mixture size, the remaining model parameters are correctly estimated.

4.2 MIXTURES OF DISTRIBUTIONS

Here we considered three gamma mixtures:

$$\begin{aligned} f_1(x) &= .25\mathcal{G}(200, 100) + .25\mathcal{G}(400, 100) + .25\mathcal{G}(600, 100) + .25\mathcal{G}(800, 100) \\ f_2(x) &= .1\mathcal{G}(9, 27) + .9\mathcal{G}(90, 270) \\ f_3(x) &= .2\mathcal{G}(40, 20) + .6\mathcal{G}(6, 1) + .2\mathcal{G}(200, 20). \end{aligned}$$

The first of these mixtures has four well-separated components with means 2, 4, 6, and 8, respectively; the second is a mixture of two gammas with equal means but different variances and weights; and the third is a mixture of a rather diffuse gamma distribution with mean 6 and two lower weighted distributions with smaller variances and means of 2 and 10, respectively.

Samples of size 400 were generated from each of the three densities. Here we compare the true data-generating densities with the predictive densities given the sampled data,

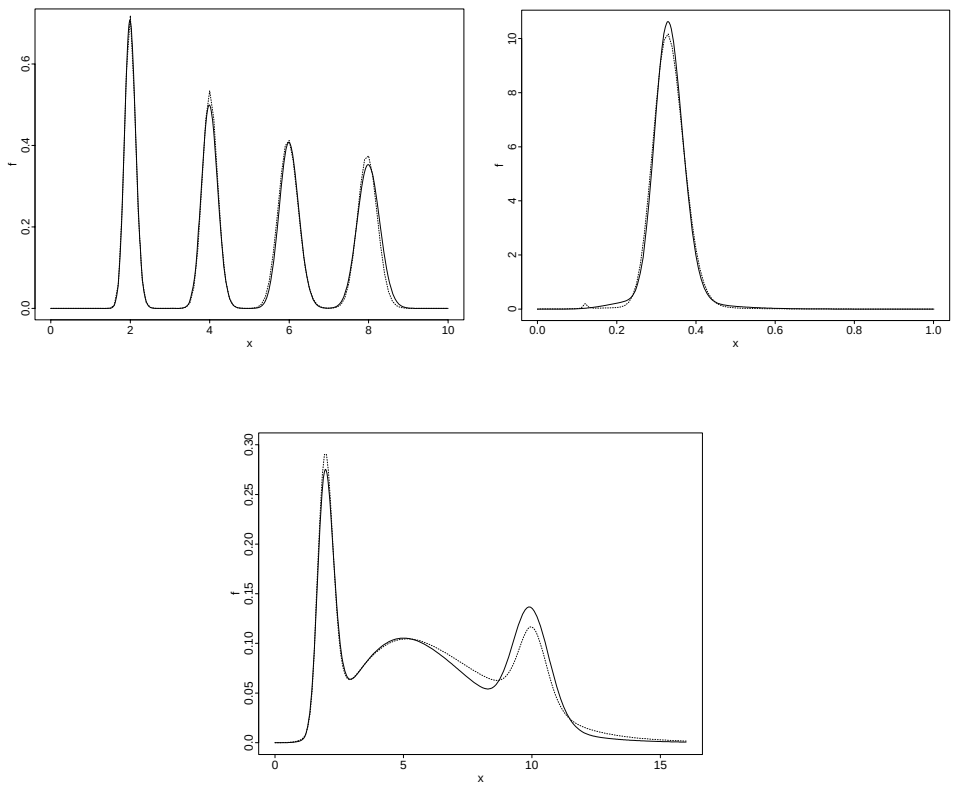


Figure 3. Generating (—) and estimated (· · ·) density functions for three gamma mixture models.

as estimated from the MCMC algorithm. The same prior distribution structure as in the previous example was used in all cases.

Figure 3 plots the true generating density (solid line) and the estimated density function (dotted line) for each of the three mixture models.

We see that the estimated and generating densities are very similar even in the third, more critical case. Note also that, conditional on the correct mixture sizes, in cases one and three, the parameters are estimated correctly; that is, the posterior median values for $\boldsymbol{\mu}$ and mean values for $\mathbf{w}$ and $\boldsymbol{\nu}$ are close to their theoretical values. Also, the method correctly predicts the number of elements in the mixture, even in the second case the posterior mode being $k = 2$ and $P(k = 1) \approx .01$ although there is some uncertainty here with larger values of k also supported. Although the two gamma distributions in the mixture have the same mean, they have very different densities and a fit using a single gamma is inadequate, so we should not be surprised by the low posterior probability for $k = 1$. The chain mixes very well in this example with the overall proportion of moves in k being around 40%. This suggests that even with the theoretical problems associated with the case of a mixture of densities with equal means, we can still carry out the density estimation in practice. Note, however, that in this example the posterior mean estimates for $\mathbf{w}$ and $\boldsymbol{\nu}$ conditional on $k = 2$ are not estimated correctly. To gain sensible posterior estimates in this example we would need to consider ordering in terms of the weights or $\boldsymbol{\nu}$.

We also carried out some sensitivity analyses, looking at the effects of small changes in the prior distributions. The effects on the density estimation problem were negligible.

Finally, we note that the density estimates given in Figure 3 are calculated using single samples only. Using averaged density estimates derived from multiple data samples obviously gives a much closer approximation.

5. AN APPLICATION TO INFERENCE FOR THE M/G/1 QUEUE

5.1 M/G/1 QUEUES

Consider a queue with Markovian arrivals (rate λ) and a general service distribution. Suppose, too, that we wish to make inference about the various parameters of the queue, in particular the traffic intensity ρ and the queue size in equilibrium (assuming $\rho < 1$).

We shall suppose that we undertake the simple experiment of observing n_a arrival times and n_s service times. Then, given the data, inference about the arrival and service parameters will be independent.

Supposing that we use a gamma prior distribution for the arrival rate, $\lambda \sim \mathcal{G}(\alpha_a, \beta_a)$, then it is straightforward to show that the posterior distribution will be

$$\lambda|\text{data} \sim \mathcal{G}(\alpha_a + n_a, \beta_a + t_a),$$

where t_a is the sum of the observed inter-arrival times. See also Armero and Bayarri (1996) and Rios Insua, Wiper, and Ruggeri (1998).

We approximate the distribution of the service time X using a mixture of gamma distributions as outlined in Sections 2 and 3. Given this model, we can estimate some of the important parameters of the queue.

The traffic intensity of the queue is defined by $\rho = \lambda \times E[X]$. It is well known that the queue is stable if and only if $\rho < 1$. See, for example, Nelson (1995). For our problem, we have

$$E[X|\mathbf{w}, \boldsymbol{\mu}] = \sum_i w_i \mu_i.$$

Thus, given the prior distribution for $\boldsymbol{\mu}$, using $\alpha = 1$ as suggested in Section 2, the posterior mean of ρ will not exist. Therefore, in this section, we use a value of $\alpha = 1.1$ which ensures existence of the mean of ρ, although the variance is still infinite.

Given data, we can estimate the posterior mean of ρ from the MCMC output using

$$E[\rho|\text{data}] \approx \frac{\alpha_a + n_a}{\beta_a + t_a} \frac{1}{N} \sum_{j=1}^{N} \sum_{i=1}^{k^{(j)}} w_i^{(j)} \mu_i^{(j)},$$

where $(\alpha_a + n_a)/(\beta_a + t_a)$ is the posterior mean of λ and $w_i^{(j)}$, $\mu_i^{(j)}$ are the weights and means for the mixture of gammas generated in the MCMC run. We can also estimate the

posterior probability that $\rho < 1$ using

$$P(\rho < 1|\text{data}) \approx \frac{1}{N} P\left(\lambda < \left(\sum_{i=1}^{k^{(j)}} w_i^{(j)} \mu_i^{(j)}\right)^{-1}\right).$$

Given that $\rho < 1$, we wish to estimate the distribution of the queue size in equilibrium. Conditional on the model parameters, it can be shown that, if we define $\pi(i) = P(i \text{ people in the queue})$, then

$$\begin{aligned} \pi(0) &= 1 - \rho \\ \pi(i) &= \pi(0)\delta_i + \sum_{j=1}^{i+1} \pi(j)\delta_{i-j+1}, \end{aligned}$$

where

$$\delta_i = \sum_{j=1}^{k} w_j \frac{\Gamma(\nu_j + i)}{i!\Gamma(\nu_j)} \left(\frac{\lambda}{\nu_j/\mu_j + \lambda}\right)^i \left(\frac{\nu_j/\mu_j}{\nu_j/\mu_j + \lambda}\right)^{\nu_j}.$$

See, for example, Nelson (1995).

Thus, we shall now simulate a sample of size N say $\lambda^{(1)}, \ldots, \lambda^{(N)}$ from the posterior distribution of λ. We can now pair off arrival and service parameters together; $(\lambda^{(1)}, k^{(1)}, \mathbf{w}^{(1)}, \boldsymbol{\mu}^{(1)}, \boldsymbol{\nu}^{(1)}), \ldots, (\lambda^{(N)}, k^{(N)}, \mathbf{w}^{(N)}, \boldsymbol{\mu}^{(N)}, \boldsymbol{\nu}^{(N)})$ which allows us to define the set $\rho^{(i)} = \lambda^{(i)} \sum_{j=1}^{k^{(i)}} w_j^{(i)} \mu_j^{(i)}$. For those $\rho^{(i)} < 1$, we can estimate the equilibrium probabilities using the algorithm above with the simulated parameters:

$$\pi(j) \approx \frac{1}{R} \sum_{i:\rho^{(i)}<1} \pi\left(j|\lambda^{(i)}, k^{(i)}, \ldots, \boldsymbol{\nu}^{(i)}\right),$$

where R is the number of $\rho^{(i)} < 1$.

5.2 CNR-IAMI E-MAIL DATA

We consider data about all the e-mail messages received at CNR-IAMI, Milan, in the period from May 5, 1997, to May 16, 1997, 9:00–18:00, Saturdays and Sundays excluded.

The arrival data are the interarrival times (minutes) of 203 E-mail messages with mean interarrival time approximately equal to 450 seconds. Fitting this data using mixtures of gammas suggested that a single, exponential distribution was an appropriate model for the arrivals. Given this model and a noninformative prior for the arrival rate λ lead to a $\mathcal{G}(202, 92461)$ posterior distribution for λ.

We assumed that service (processing) times were proportional to size of the E-mail. The large majority of the E-mails processed were very small (fewer than 20 lines), though there were a number of large outliers in this data. Thus, the service distribution is both multimodal and highly skewed. This is probably due to different kinds of messages received: usually, conversational E-mails tend to be much smaller than postscript file E-mails. The mean of the service data is $\bar{x} = 5.00$. Figure 4 shows a histogram of the logged service times

and a histogram of the original data (ignoring the largest three outliers). In each case, the estimated density function using the gamma mixture model for the original data has been superimposed.

Figure 4(a) illustrates the overall skewness of the data and the outliers and Figure 4(b) shows that this large degree of skewness persists even in the smaller E-mails. The fit of the gamma mixture model appears reasonable.

In this case there was a lot of posterior uncertainty about the number of terms in the mixture model for the service data with the posterior mode for the mixture size being 4, having probability approximately equal to .25, but with values of k between 3 and 6 all having posterior probabilities of .1 or more.

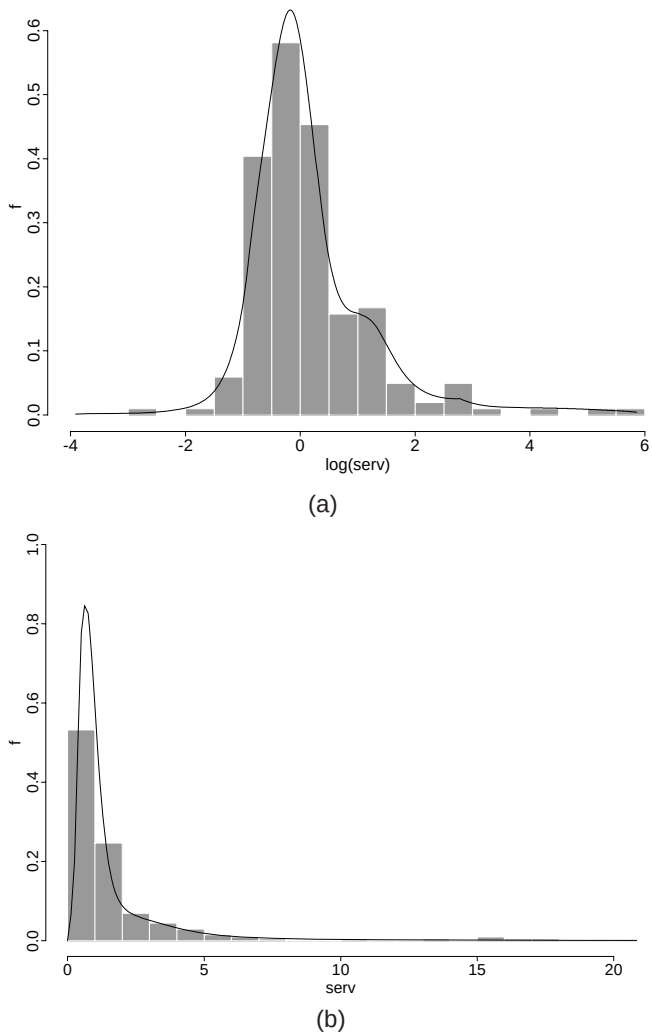


Figure 4. Histograms of logged data (a) and of the original data (b) minus largest outliers with estimated densities superimposed.

Table 1. Estimated Queueing Parameters Given Different Posterior Distributions for λ

$E[\lambda \mid \text{data}]$	$\frac{202}{9246.1}$	$\frac{202}{4623.1}$	$\frac{202}{924.61}$
$E[\rho \mid \text{data}]$	.12	.24	1.21
$P(\rho < 1 \mid \text{data})$	.999	.997	.445
$E[\rho \mid \rho < 1, \text{data}]$	.12	.24	.79

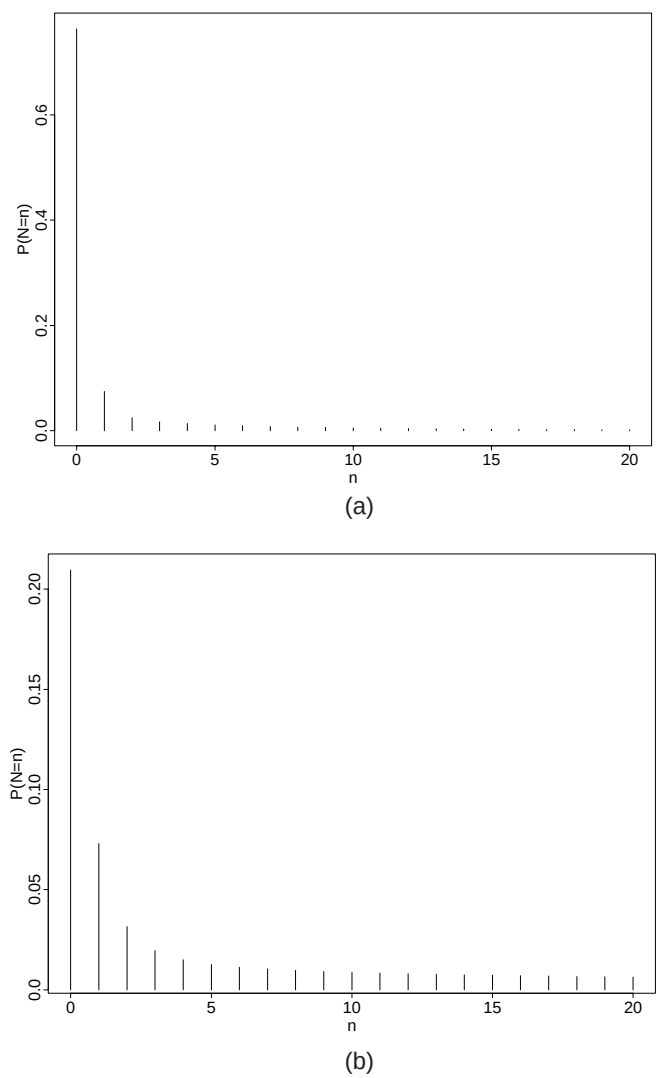


Figure 5. Distributions of queue size in equilibrium for $E[\lambda] = 202/4623.1$ and $E[\lambda] = 202/924.61$.

Given these data, the posterior probability that the traffic intensity is less than one was estimated to be equal to unity to 3 decimal places and the expected value of ρ was estimated as .01. The probability that the queue is empty in equilibrium was estimated to be .99.

We can also consider the effects of different arrival rates on the queueing estimates. In Table 1 we modify the second parameter of the arrival distribution to allow us to simulate queues with different traffic intensities.

Figure 5 gives the posterior probabilities for the distribution of the queue size in equilibrium in the latter two cases shown in Table 1. The limiting distributions in the two different cases have very different forms.

Finally, we also used these data to explore sensitivity to our election of prior distributions. We carried out various changes to the prior parameters: for the distribution on $\mathbf{w}$ taking $.5 < \phi < 1$, for $\boldsymbol{\mu}$ letting α get close to 1 and taking $.1 < \beta < 1$ and for $\boldsymbol{\nu}$ setting $.001 < \theta < .1$. We found that there was, in general, very little sensitivity in the queueing estimates to these changes. For example, analyzing the most unstable queue with $E[\lambda|\text{data}] = 202/924.61$, we found that $P(\rho < 1|\text{data})$ varied between .43 and .46. Similarly, the posterior mean $E[\rho|\text{data}]$ changed very little, increasing very slowly as α got very close to 1. In fact, much larger MCMC samples were needed (5,000,000 iterations in equilibrium) to see the increase. Thus, it appears from this example that inference is relatively insensitive to the choice of prior distribution.

6. CONCLUSIONS

This article carries out Bayesian inference using a gamma mixture model. This approach allows us to model both real and simulated datasets. The use of gamma mixtures is sound from both a theoretical and practical point of view, since all distributions on the positive reals can be approximated by them and features such as skewness can be better modeled by asymmetric distributions. Implementation of the method is straightforward, but attention should be paid in tuning the initial parameter values. This is one of the issues to be considered.

We note that, from our practical experience, the acceptance rates for the reversible jump algorithm can vary wildly with different datasets. We have had acceptance rates as low as 2% and as high as 45%. We note that the algorithm we use for splitting and joining terms in the mixture is far from unique and it may well be that with alternative election of the various generators used for this algorithm, we could find more consistent acceptance rates.

In analyzing queueing data, it would be more natural to consider mixtures of Erlang distributions (gamma distributions with integer shape parameter) than gamma distributions, as these have a natural interpretation in this context; see, for example, Nelson (1995). A similar algorithm to that used here would be appropriate for Erlang mixtures, and this presents no problems conditional on the mixture size k. However, there are technical problems to be addressed with the reversible jump algorithm in this case. In particular, the preservation of second moments in the combine step of the reversible jump algorithm is more complicated to achieve and thus is likely to lead to a lower mixing rate over different values of the mixture size. Work is currently in progress on this model.

Furthermore, we might consider some reparameterization to facilitate the use of non-

informative priors, in the spirit of Gruet, Philippe, and Robert (1999), but we believe that this is unlikely to substantially change the produced density estimates.

Finally, we might compare density estimates using both gamma and normal mixtures, although this is beyond the scope of this article.

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